

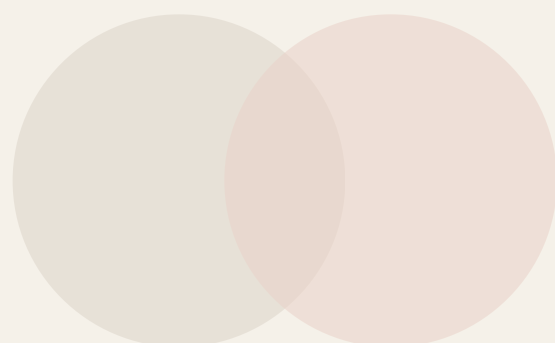
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DR. JARKKO MOILANEN

# DATA QUALITY IS NOT DATA PRODUCT QUALITY

## Two Quality Systems, One Data Product

A fact-based framework for separating data quality from the quality of the complete data product.



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# Abstract

Data quality and data product quality are often treated as synonyms. They should not be. They assess different objects and answer different management questions.

ISO/IEC 25012 defines a data-quality model for structured data, while ISO/IEC 25024 defines measures for assessing those characteristics across the data lifecycle. <sup>S01 S02</sup> A data product is a broader managed offering. Data-mesh product thinking includes data, metadata, code, interfaces, infrastructure, policy enforcement, observability and ownership. <sup>S04</sup> The Open Data Contract Standard makes the distinction explicit by placing data quality beside schema, support, roles, service levels and infrastructure. <sup>S06</sup>

**Data quality evaluates whether the data are fit for their intended use.**

**Data product quality evaluates whether the complete managed offering delivers useful, trustworthy, reliable, secure and supportable outcomes to its consumers.**

The concepts overlap. Broad data-quality frameworks already include accessibility, interpretability and system-dependent characteristics. <sup>S01 S03</sup> The reliable distinction is therefore not a simplistic boundary around individual metrics. It is the object under assessment: the data versus the complete end-to-end data product that delivers and supports those data.

The operational consequence is material. An organization might have 99.9 percent accurate records and still have a poor data product because consumers cannot find it, understand it, gain access to it, rely on availability, survive a breaking change or obtain support. The reverse also occurs. A polished, well-documented API might reliably deliver inaccurate data.

**Central recommendation.** Never use a single “data quality score” as the health score for a data product. Maintain two linked scorecards. One measures the properties of the data. The other measures the fitness of the data product as a managed offering.

# The Mistake Is Understandable

People with a strong background in traditional data management have good reasons to treat data quality as the dominant quality discipline. For decades, the practical problems were accuracy, completeness, consistency, timeliness, semantics, lineage and access. Mature data-quality frameworks also extended well beyond simple correctness.

Wang and Strong found that consumers care about intrinsic, contextual, representational and accessibility quality. Their framework includes relevance, timeliness, completeness, interpretability, ease of understanding, accessibility and access security. <sup>S03</sup> ISO/IEC 25012 similarly distinguishes inherent and system-dependent quality. <sup>S01</sup>

That history explains the confusion. If a practitioner has spent ten years improving data quality, many of the concerns now discussed under data product quality look familiar. Some of them are familiar. The mistake is assuming that overlap makes the two concepts identical.

A product lens changes the unit of analysis. It asks whether a consumer can repeatedly find, understand, access, use and depend on a managed offering. Data correctness remains central, but it becomes one assurance domain inside a larger operating responsibility.

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**The decisive distinction is not “data versus technology.” It is data quality versus the quality of the complete data product.**

# Define the Two Quality Objects

## 2.1 Data quality

ISO/IEC 25012 defines a general quality model for structured data and supports the definition of requirements, measures, evaluation and improvement. ISO/IEC 25024 turns that model into measurable characteristics and leaves acceptable thresholds to system context and user requirements. <sup>S01 S02</sup>

**Data quality is the degree to which the characteristics of data satisfy the requirements of a specified use and its consumers.**

## 2.2 Data product quality

A data product requires a different unit of analysis. Dehghani describes the product as a composition of code, data, metadata and infrastructure. The implementation includes pipelines, APIs, schemas, observability, access control, provenance and compliance policies. Metadata includes semantic definitions and quality metrics. <sup>S04</sup>

The Open Data Product Standard reinforces that broader view through product information, context, management ports, support and communications, teams and authoritative definitions. Its companion Open Data Contract Standard states that the producer-consumer agreement covers schema, data quality, support, team, roles, service levels and infrastructure. <sup>S05 S06</sup>

**Data product quality is the degree to which the complete managed data offering satisfies consumer, business, operational, security, governance and lifecycle requirements.**

## 2.3 Data quality is necessary, but not sufficient

ISO/IEC 25010 is useful only as a structural analogy. It assesses the quality of the whole ICT product rather than one type of content inside it. It is not a dedicated data-product-quality standard. <sup>S08</sup> The practical lesson is narrower: evaluating the data alone does not evaluate the complete data product.

# Why Overlap Does Not Mean Equivalence

The distinction becomes clear when the assessment question changes. The same organization can have strong data quality and weak data product quality, or the reverse.

| Question                              | Data quality                                           | Data product quality                                                                                                                |
|---------------------------------------|--------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| <b>Primary object</b>                 | Values, records, datasets and their semantics          | The end-to-end data product offered to consumers                                                                                    |
| <b>Core question</b>                  | Are these data fit for this use?                       | Does this data product reliably deliver value to its consumers?                                                                     |
| <b>Typical failure</b>                | Wrong, missing, stale, inconsistent or misleading data | Unavailable service, hard-to-find product, poor documentation, access friction, breaking interface, weak support, uncontrolled cost |
| <b>Data correctness</b>               | Central                                                | Central, but only one quality domain                                                                                                |
| <b>Availability</b>                   | Often assessed as accessibility or system-dependent DQ | Explicit service concern for the data product                                                                                       |
| <b>Interface stability</b>            | Usually outside narrow DQ                              | Core data product concern                                                                                                           |
| <b>Documentation</b>                  | Supports interpretability                              | Core onboarding and usability concern                                                                                               |
| <b>Security/privacy</b>               | Sometimes included as DQ access security               | Full control covering data, interface, users and infrastructure                                                                     |
| <b>Versioning and deprecation</b>     | Rarely a primary DQ dimension                          | Core lifecycle responsibility                                                                                                       |
| <b>Support model</b>                  | Outside most DQ measurement                            | Data product quality concern                                                                                                        |
| <b>Adoption and consumer outcomes</b> | Context for determining fitness                        | Direct data product outcome                                                                                                         |
| <b>Primary accountability</b>         | Shared by domain owners, stewards and engineering      | Data product owner and product team, with domain and governance partners                                                            |

This synthesis follows the separation between data-quality measurement and whole-product assessment, Dehghani's data-product composition, and the explicit separation of DQ from service levels, support and infrastructure in ODCS.

S01 S02 S04 S06

# The Diagnostic Tests

A practical boundary test starts by holding one quality domain constant and asking whether the consumer can still fail.

## Test A: assume perfect data

Assume every record is accurate, complete and current. Could the consumer still fail because the product is hard to find, access, interpret, operate or support?

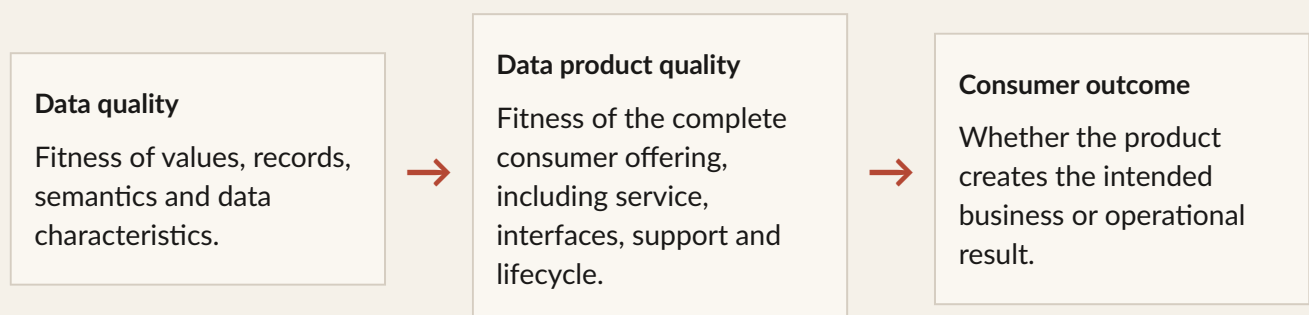
## Test B: assume a polished product

Assume the data product has excellent uptime, clear documentation, stable interfaces and fast support. Would incorrect records still harm the consumer?

If Test A is positive, the remaining failures belong to data product quality. If Test B is positive, the data product may operate well while still carrying poor data quality.

## 4.1 Four possible states

| Data quality | Data product quality | Interpretation                                                              |
|--------------|----------------------|-----------------------------------------------------------------------------|
| High         | High                 | Desired state. Trustworthy data delivered through a dependable data product |
| High         | Low                  | Good data trapped in a poor data product                                    |
| Low          | High                 | Well-engineered delivery of unreliable data                                 |
| Low          | Low                  | Neither the data nor the offering is fit for use                            |



The product boundary is broader, but it does not erase the data-quality boundary. The two quality systems should connect through the product contract and consumer outcome.

# Two Scorecards, Not One

Different quality dimensions matter differently to different use cases. ISO/IEC 25024 does not prescribe one universal passing grade. <sup>S02</sup> That argues against a single enterprise “DQ percentage” and, even more strongly, against using that percentage as the health score for a data product.

## 5.1 Data-quality measures

| Quality domain           | Typical metric                        | Example calculation                                                |
|--------------------------|---------------------------------------|--------------------------------------------------------------------|
| Accuracy                 | Error rate or verified accuracy       | Correct verified values / assessed values                          |
| Completeness             | Required-data completion              | Populated required values / expected required values               |
| Validity                 | Conformance to defined rules          | Values passing domain/schema rule / assessed values                |
| Consistency              | Cross-field or cross-source agreement | Reconciled items / compared items                                  |
| Timeliness               | Age relative to business need         | Records inside required age window / records                       |
| Uniqueness               | Duplicate rate                        | Duplicate entities / entities assessed                             |
| Referential integrity    | Valid relationship rate               | Resolved references / references                                   |
| Representational clarity | Defined and interpretable fields      | Fields with approved semantics/unit definitions / published fields |
| Traceability             | Lineage coverage                      | Critical fields with traceable origin / critical fields            |

## 5.2 Data-product-quality measures

| Data-product-quality domain | Typical KPI or SLO                                 | What it detects                                          |
|-----------------------------|----------------------------------------------------|----------------------------------------------------------|
| Embedded data quality       | Critical DQ SLO attainment                         | Whether product outputs remain fit for use               |
| Reliability                 | Availability, successful publication rate          | Whether consumers receive the product when promised      |
| Freshness service           | On-time delivery cycles                            | Whether the service executes its publication commitment  |
| Performance                 | Query/API latency under agreed workload            | Whether consumption is operationally practical           |
| Discoverability             | Time from search to correct product identification | Whether consumers find the right product                 |
| Access experience           | Time from approved request to usable access        | Whether governance creates unnecessary consumption delay |

| Data-product-quality domain | Typical KPI or SLO                               | What it detects                                               |
|-----------------------------|--------------------------------------------------|---------------------------------------------------------------|
| Understandability           | Required semantic/documentation coverage         | Whether consumers understand meaning, units and limitations   |
| Contract stability          | Breaking-change incidents                        | Whether producers protect consumers from uncontrolled change  |
| Supportability              | Incident acknowledgement and resolution time     | Whether consumers receive operational support                 |
| Security and privacy        | Control compliance, access violations, incidents | Whether delivery respects security and regulatory obligations |
| Adoption                    | Active intended consumers and recurring use      | Whether the product gets used for its intended purpose        |
| Consumer outcomes           | Target use cases meeting acceptance criteria     | Whether use creates the intended business outcome             |
| Lifecycle management        | Version/deprecation compliance                   | Whether change and retirement occur predictably               |

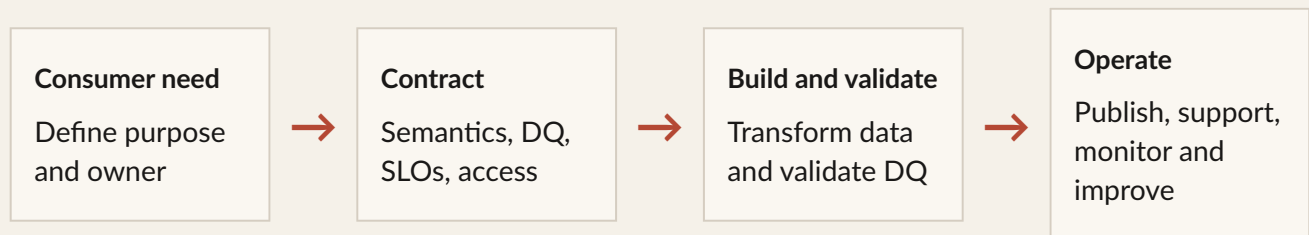
One metric sometimes appears in both scorecards because the measured obligation differs. “95 percent of events are present within five minutes of occurrence” measures data timeliness. “99.9 percent of scheduled publications complete before 07:00 UTC” measures service reliability. The clocks are related, but the responsibilities are different.

**Do not average unlike obligations into one percentage.** A data product with 100 percent record accuracy and zero availability is unusable. A mathematical average can hide a hard failure.

# Lifecycle and Accountability

Data-quality management runs through acquisition, transformation, storage, integration and consumption. Data-product management adds consumer discovery, interface design, access, support, operations, versioning and retirement.

S02 S04



The key governance point is span. DQ controls sit inside the lifecycle. Data product quality accountability runs from consumer need through retirement.

## 6.1 Ownership is shared, accountability is not vague

Product accountability belongs with a role that owns consumer outcomes and the complete service. Stewardship owns or co-owns meaning, definitions and quality rules. Engineering operationalizes those rules. Platform functions provide reusable reliability and observability capabilities. Governance specialists set and assure regulatory controls.

| Responsibility area                 | Primary responsibility          | Key partners                               |
|-------------------------------------|---------------------------------|--------------------------------------------|
| Product purpose and outcomes        | Data product owner              | Domain, consumers, governance              |
| Semantics and definitions           | Data steward                    | Data product owner, engineering, consumers |
| DQ rules and thresholds             | Data steward + data engineering | Data product owner, governance             |
| Availability and performance        | Engineering + platform/SRE      | Data product owner, security/risk          |
| Access and security                 | Engineering + security/risk     | Data product owner, steward                |
| Contracts, versioning and lifecycle | Data product owner              | Steward, engineering, platform, consumers  |

The full RACI is retained in Appendix D.

# Governance and Tooling

A team that treats DQ and data product quality as synonyms tends to concentrate controls around profiling and validation. Those controls are necessary. They leave major product failure modes uncovered.

| Control area | Data-quality practice                             | Data-product-quality practice                                                        |
|--------------|---------------------------------------------------|--------------------------------------------------------------------------------------|
| Requirements | Define DQ dimensions and thresholds by use case   | Define consumers, purpose, outcomes, interfaces and whole-data-product SLOs          |
| Design       | Semantic models, validation rules, reference data | Output ports, contracts, access model, support model, version strategy               |
| Build        | Schema validation, rule testing, reconciliation   | CI/CD, contract compatibility, documentation, policy enforcement                     |
| Release      | DQ gate                                           | DQ gate plus security, interface, metadata, reliability and support gates            |
| Operate      | Monitor anomalies and quality thresholds          | Monitor DQ, availability, latency, freshness, incidents, usage and cost              |
| Govern       | Stewardship, issue management, definitions        | Product portfolio, ownership, contracts, lifecycle and deprecation                   |
| Improve      | Root cause analysis of bad data                   | Consumer feedback, incident learning, adoption/value review and data product roadmap |

## 7.1 Tooling should mirror the split

DQ tooling should support profiling, rule execution, schema and domain validation, reconciliation, duplicate detection, anomaly detection, reference-data checks, lineage and DQ incident management.

Data-product tooling should add catalog and discovery, contracts, versioning, interface management, observability, service SLOs, access control, documentation, support, incident management and usage analytics. A data product is broader than the dataset and the validation suite that protects it. <sup>S04 S05 S06</sup>

# Compliance Shows the Boundary

Regulation and sector standards expose the danger of conflation because obligations attach to different objects.

GDPR Article 5 requires personal data to be accurate and, where necessary, kept up to date. The same article separately requires security against unauthorized processing and accidental loss, destruction or damage, and assigns accountability to the controller. Accuracy is a data-quality obligation. Security and accountability extend beyond it. <sup>S09</sup>

The EU AI Act follows the same pattern for high-risk AI. Article 10 sets requirements for training, validation and testing data, including relevance, representativeness, completeness and error control, while the broader system remains subject to provenance, governance, documentation, monitoring and lifecycle controls. <sup>S10</sup>

Financial services make the business-risk connection explicit. BCBS 239 emerged from weaknesses exposed by the financial crisis, including the inability of some banks to aggregate risk exposures fully, quickly and accurately. Accuracy alone was insufficient without aggregation and reporting capability. <sup>S11</sup>

Geospatial practice provides one of the cleanest standards-based separations. ISO 19157 treats data quality as evidence for assessing how well a dataset conforms to a broader data-product specification. <sup>S07</sup>

| Context                      | DQ emphasis                                                                         | Additional data-product-quality emphasis                                 |
|------------------------------|-------------------------------------------------------------------------------------|--------------------------------------------------------------------------|
| General enterprise analytics | Accuracy, completeness, consistency, timeliness                                     | Discovery, semantics, access, reliability, support, versioning           |
| Financial risk               | Accuracy, completeness, aggregation integrity and timeliness                        | Rapid aggregation, reliable reporting, governance and stress readiness   |
| Personal data                | Accuracy and currentness                                                            | Lawful access, security, accountability and lifecycle controls           |
| High-risk AI                 | Relevance, representativeness, completeness, error control, bias-related properties | Provenance, governance, monitoring, documentation and risk controls      |
| Geospatial                   | Positional/thematic/temporal dataset quality against specification                  | Product specification, metadata, delivery and intended-use documentation |

A “DQ passed” result does not establish data-product compliance. Some obligations attach to the data. Others attach to processing, access, interfaces, organizations and operational systems.

# Failure Cases Expose It

## 9.1 Accurate source results, failed reporting product

Public Health England reported that 15,841 positive COVID-19 cases from September 25 through October 2, 2020 had not appeared in daily reported case figures. Affected people had received their individual test results normally. The failure occurred in the automated data-load process used for reporting dashboards and contact tracing. Some files exceeded the maximum size accepted by the central loading process. <sup>S12</sup>

This was not simply “bad source data.” The downstream reporting product failed, producing incomplete and late aggregate reporting.

| Failure                                       | Classification                                           |
|-----------------------------------------------|----------------------------------------------------------|
| Positive cases missing from reporting dataset | DQ completeness failure                                  |
| Cases appearing after intended reporting time | DQ timeliness failure                                    |
| File-size constraint not handled end to end   | Data product robustness failure                          |
| Automated transfer failed                     | Data product reliability failure                         |
| Detection occurred after backlog              | Data product observability/control failure               |
| Need for end-to-end review                    | Data product lifecycle and operational assurance failure |

Calling the whole incident a “data quality problem” is not wrong, but it is incomplete. A remediation limited to validation rules would miss the product mechanism that created the DQ defect.

## 9.2 Accurate data that consumers still could not use safely

Wang and Strong describe an investment firm whose managers initially believed their databases had perfect quality because the data were accurate. Consumers still faced missing source context, insufficient timeliness, and implicit currencies and units. <sup>S03</sup>

| Observed issue            | DQ view                  | Data product view                   |
|---------------------------|--------------------------|-------------------------------------|
| Source reputation unclear | Believability/provenance | Metadata and trust                  |
| Data arrive too late      | Timeliness               | Freshness SLO and delivery          |
| Currency implicit         | Interpretability         | Contract and semantic documentation |
| Units implicit            | Representational quality | Interface and documentation quality |

A single accuracy KPI would have shown green while consumers faced material usage risk.

## 9.3 Correct separation: Landsat

A USGS Landsat Level-1 product combines calibrated image data with metadata and radiometric coefficients needed to interpret the stored values. Data quality remains separately assessable inside the product. <sup>S13 S07</sup> The lesson generalizes: a high-quality data product packages enough semantics, metadata, constraints, interfaces and operating commitments for consumers to use the data correctly.

# Operating Model: SLAs and KPIs

Use three levels of measurement. The first measures data properties. The second measures the data product service. The third measures outcomes. The thresholds below are synthetic examples, not universal standards. ISO/IEC 25024 explicitly ties acceptable values to context and user needs. <sup>S02</sup>

| Layer                     | Example commitment                                                   |
|---------------------------|----------------------------------------------------------------------|
| DQ, completeness          | Critical identifiers populated in at least 99.9% of eligible records |
| DQ, validity              | At least 99.5% of critical fields satisfy agreed domain rules        |
| DQ, reconciliation        | Daily financial control difference no greater than 0.1%              |
| DQ, event timeliness      | 95% of events available within 15 minutes of source event            |
| Data product availability | At least 99.9% monthly availability for supported output             |
| Data product publication  | At least 99.5% of scheduled runs complete by agreed deadline         |
| Data product latency      | P95 response below 5 seconds for agreed reference workload           |
| Contract stability        | Zero unannounced breaking changes                                    |
| Change management         | At least 30 days' notice for agreed breaking-change classes          |
| Access                    | P90 approved access provisioned within one business day              |
| Support                   | Severity-1 acknowledgement within 15 minutes                         |
| Recovery                  | Severity-1 service restored within two hours                         |
| Documentation             | 100% of critical fields have approved meaning, type and unit         |
| Lineage                   | 100% of critical outputs traceable to governed upstream sources      |
| Outcome                   | Agreed target use cases meet data-product-specific success criteria  |

## 10.1 Executive health view

A product health review should display the components separately rather than hiding them in one composite score.

| Dimension         | Status | Trend     | SLO breach? | Owner          |
|-------------------|--------|-----------|-------------|----------------|
| Data correctness  | Green  | Stable    | No          | Steward/domain |
| Data completeness | Amber  | Declining | Yes         | Steward/domain |

| Dimension           | Status | Trend     | SLO breach? | Owner                  |
|---------------------|--------|-----------|-------------|------------------------|
| Freshness           | Green  | Stable    | No          | Engineering            |
| Availability        | Green  | Stable    | No          | Engineering/platform   |
| Contract stability  | Red    | Declining | Yes         | Data product owner     |
| Access lead time    | Amber  | Improving | No          | Product/governance     |
| Security/compliance | Green  | Stable    | No          | Security/risk          |
| Adoption            | Green  | Growing   | N/A         | Data product owner     |
| Consumer outcome    | Amber  | Stable    | N/A         | Product owner/business |

# Release Readiness

A data product should pass a data-product-level readiness gate rather than only a DQ gate. ODPS and ODCS provide a structural basis through their coverage of product information, contracts, management, support, teams, quality and service levels. <sup>S05 S06</sup>

| Gate             | Required evidence                                                |
|------------------|------------------------------------------------------------------|
| Purpose          | Named consumers, intended uses, limitations and business outcome |
| Ownership        | Accountable data product owner and operational contacts          |
| Semantics        | Definitions, units, business meaning and authoritative sources   |
| Interface        | Documented output ports, schema and access method                |
| Data quality     | Rules, thresholds, measurement method and escalation path        |
| Reliability      | Availability, freshness, performance and recovery SLOs           |
| Observability    | Monitoring for pipeline, DQ and data product SLO failures        |
| Security/privacy | Classification, access policy, controls and regulatory review    |
| Discoverability  | Catalog entry and searchable product metadata                    |
| Support          | Support channel, severity definitions and response expectations  |
| Change           | Versioning, compatibility and deprecation policy                 |
| Lineage          | Upstream and downstream critical dependencies                    |
| Cost             | Data product operating cost and material consumption costs       |
| Value            | Adoption and consumer-outcome measures                           |
| Retirement       | Consumer notification and migration process                      |

**Passing every DQ test should be necessary for critical data products. It should not constitute the data product release decision by itself.**

# A 90-Day Remediation Path

A migration does not require replacing the data-quality program. It requires placing that program inside a larger data-product-management system.

| Period        | Action                       | Deliverable                                                                                               |
|---------------|------------------------------|-----------------------------------------------------------------------------------------------------------|
| Days 1-30     | Establish definitions        | Approved glossary for dataset, DQ, data product, data contract and data product quality                   |
| Days 1-30     | Inventory critical products  | Consumer, owner, purpose, outputs and dependencies for each                                               |
| Days 1-30     | Split current health score   | Existing DQ metrics retained, data product gaps recorded separately                                       |
| Days 31-60    | Assign accountability        | Data product owner, steward, engineering and governance responsibilities                                  |
| Days 31-60    | Introduce product contracts  | Schema, semantics, DQ thresholds, SLA/SLO, support and security commitments                               |
| Days 31-60    | Establish dual dashboards    | DQ scorecard plus data-product/service/outcome scorecard                                                  |
| Days 61-90    | Establish release gates      | Data product readiness criteria incorporated into delivery workflow                                       |
| Days 61-90    | Separate incident taxonomy   | DQ incident, service incident, contract incident, security incident                                       |
| Days 61-90    | Introduce lifecycle controls | Versioning, consumer notification, deprecation and retirement process                                     |
| After 90 days | Scale by risk                | Apply stronger requirements to products with material regulatory, financial, safety or operational impact |

## 12.1 Remove three anti-patterns

First, retire statements such as “this product has 98% quality” unless the score specifies what it measures. Acceptable data-quality values depend on context. <sup>S02</sup>

Second, stop treating a successful DQ test suite as a production-readiness certificate. The PHE incident shows why. Source results existed, yet an end-to-end transfer limitation still created incomplete and delayed reporting. <sup>S12</sup>

Third, stop assigning “quality” to the steward alone. Modern data-product architecture places code, interfaces, metadata, infrastructure and operational controls inside the product responsibility. <sup>S04</sup>

# Stakeholder Primer

**Data quality asks whether you should trust the data.**

**Data product quality asks whether you should depend on the data product that delivers those data.**

The wording is intentionally simplified. Broad DQ frameworks include accessibility and system-related characteristics. The point is to give teams a practical operating distinction without erasing the overlap.

| Stakeholder                            | What matters most                                                                                        | What not to confuse                                                      |
|----------------------------------------|----------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------|
| Engineers                              | Implement DQ rules, reliable pipelines, contracts, interfaces, observability, security and compatibility | Passing data tests does not prove the service is reliable                |
| Data stewards                          | Define semantics, critical data elements, quality rules, thresholds and issue processes                  | Stewardship of DQ does not equal ownership of the whole data product     |
| Product managers / data product owners | Define consumers, outcomes, interfaces, SLOs, support, roadmap, adoption and lifecycle                   | Data product health is broader than a DQ dashboard                       |
| Executives                             | Fund accountable data products and monitor value, reliability, risk and DQ separately                    | A single enterprise DQ percentage does not represent data-product health |
| Risk/compliance                        | Map obligations to data, processing, access, operations and lifecycle                                    | Data accuracy does not establish regulatory compliance                   |
| Consumers                              | State required fitness, freshness, semantics, reliability and change tolerance                           | Certified data does not guarantee a usable consumption experience        |

# Final Position

Data quality and data product quality are distinct but overlapping quality domains. In an operating model, data quality is a required assurance domain within data product quality, but broad DQ frameworks also include accessibility, representation and system-dependent properties. The reliable way to distinguish the concepts is by the object and responsibility being assessed.

The evidence converges from different directions. Wang and Strong show that data quality already extends well beyond accuracy. <sup>S03</sup> ISO separates data-quality models and measures from broader whole-product models. <sup>S01 S02 S08</sup> ISO 19157 places dataset quality against a broader product specification. <sup>S07</sup> Modern data-product architecture treats data as one component beside code, metadata, interfaces, infrastructure and policy. <sup>S04</sup> ODCS explicitly places data quality beside service levels, support, roles and infrastructure in the producer-consumer contract. <sup>S06</sup>

## 14.1 Five operating rules

- Assess the dataset and the data product separately.
- Put measurable DQ requirements inside the product contract.
- Give the data product an accountable owner beyond stewardship.
- Measure service, usability, security, lifecycle and consumer outcomes alongside DQ.
- Never allow an aggregate DQ score to stand in for data-product health.

**Data quality is necessary for a high-quality data product. It is not sufficient.**

That separation preserves the mature discipline of data quality while adding the product responsibilities required for data that people and systems must find, understand, access, trust and depend upon.

# Appendix A. Core Comparison Matrix

| Question                       | Data quality                                           | Data product quality                                                                                                                |
|--------------------------------|--------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| Primary object                 | Values, records, datasets and their semantics          | The end-to-end data product offered to consumers                                                                                    |
| Core question                  | Are these data fit for this use?                       | Does this data product reliably deliver value to its consumers?                                                                     |
| Typical failure                | Wrong, missing, stale, inconsistent or misleading data | Unavailable service, hard-to-find product, poor documentation, access friction, breaking interface, weak support, uncontrolled cost |
| Data correctness               | Central                                                | Central, but only one quality domain                                                                                                |
| Availability                   | Often assessed as accessibility or system-dependent DQ | Explicit service concern for the data product                                                                                       |
| Interface stability            | Usually outside narrow DQ                              | Core data product concern                                                                                                           |
| Documentation                  | Supports interpretability                              | Core onboarding and usability concern                                                                                               |
| Security/privacy               | Sometimes included as DQ access security               | Full control covering data, interface, users and infrastructure                                                                     |
| Versioning and deprecation     | Rarely a primary DQ dimension                          | Core lifecycle responsibility                                                                                                       |
| Support model                  | Outside most DQ measurement                            | Data product quality concern                                                                                                        |
| Adoption and consumer outcomes | Context for determining fitness                        | Direct data product outcome                                                                                                         |
| Primary accountability         | Shared by domain owners, stewards and engineering      | Data product owner and product team, with domain and governance partners                                                            |

# Appendix B. Data Quality Metrics

| Quality domain           | Typical metric                        | Example calculation                                                |
|--------------------------|---------------------------------------|--------------------------------------------------------------------|
| Accuracy                 | Error rate or verified accuracy       | Correct verified values / assessed values                          |
| Completeness             | Required-data completion              | Populated required values / expected required values               |
| Validity                 | Conformance to defined rules          | Values passing domain/schema rule / assessed values                |
| Consistency              | Cross-field or cross-source agreement | Reconciled items / compared items                                  |
| Timeliness               | Age relative to business need         | Records inside required age window / records                       |
| Uniqueness               | Duplicate rate                        | Duplicate entities / entities assessed                             |
| Referential integrity    | Valid relationship rate               | Resolved references / references                                   |
| Representational clarity | Defined and interpretable fields      | Fields with approved semantics/unit definitions / published fields |
| Traceability             | Lineage coverage                      | Critical fields with traceable origin / critical fields            |

These metrics represent common operationalizations of fitness-for-use concepts and the measurement approach in ISO/IEC 25024. <sup>S02 S03</sup>

# Appendix C. Data Product Quality Metrics

| Data-product-quality domain | Typical KPI or SLO                                 | What it detects                                               |
|-----------------------------|----------------------------------------------------|---------------------------------------------------------------|
| Embedded data quality       | Critical DQ SLO attainment                         | Whether product outputs remain fit for use                    |
| Reliability                 | Availability, successful publication rate          | Whether consumers receive the product when promised           |
| Freshness service           | On-time delivery cycles                            | Whether the service executes its publication commitment       |
| Performance                 | Query/API latency under agreed workload            | Whether consumption is operationally practical                |
| Discoverability             | Time from search to correct product identification | Whether consumers find the right product                      |
| Access experience           | Time from approved request to usable access        | Whether governance creates unnecessary consumption delay      |
| Understandability           | Required semantic/documentation coverage           | Whether consumers understand meaning, units and limitations   |
| Contract stability          | Breaking-change incidents                          | Whether producers protect consumers from uncontrolled change  |
| Supportability              | Incident acknowledgement and resolution time       | Whether consumers receive operational support                 |
| Security and privacy        | Control compliance, access violations, incidents   | Whether delivery respects security and regulatory obligations |
| Adoption                    | Active intended consumers and recurring use        | Whether the product gets used for its intended purpose        |
| Consumer outcomes           | Target use cases meeting acceptance criteria       | Whether use creates the intended business outcome             |
| Lifecycle management        | Version/deprecation compliance                     | Whether change and retirement occur predictably               |

# Appendix D. Responsibility Model

| Responsibility area                 | Primary responsibility          | Key partners                               |
|-------------------------------------|---------------------------------|--------------------------------------------|
| Product purpose and outcomes        | Data product owner              | Domain, consumers, governance              |
| Semantics and definitions           | Data steward                    | Data product owner, engineering, consumers |
| DQ rules and thresholds             | Data steward + data engineering | Data product owner, governance             |
| Availability and performance        | Engineering + platform/SRE      | Data product owner, security/risk          |
| Access and security                 | Engineering + security/risk     | Data product owner, steward                |
| Contracts, versioning and lifecycle | Data product owner              | Steward, engineering, platform, consumers  |

The full RACI is retained in Appendix D.

# Appendix E. Data Product Quality Checklist

- **Purpose.** Named consumers, intended uses, limitations and business outcome
  - **Ownership.** Accountable data product owner and operational contacts
  - **Semantics.** Definitions, units, business meaning and authoritative sources
  - **Interface.** Documented output ports, schema and access method
  - **Data quality.** Rules, thresholds, measurement method and escalation path
  - **Reliability.** Availability, freshness, performance and recovery SLOs
  - **Observability.** Monitoring for pipeline, DQ and data product SLO failures
  - **Security/privacy.** Classification, access policy, controls and regulatory review
  - **Discoverability.** Catalog entry and searchable product metadata
  - **Support.** Support channel, severity definitions and response expectations
  - **Change.** Versioning, compatibility and deprecation policy
  - **Lineage.** Upstream and downstream critical dependencies
  - **Cost.** Data product operating cost and material consumption costs
  - **Value.** Adoption and consumer-outcome measures
  - **Retirement.** Consumer notification and migration process
- 

## Dual-scorecard check

- Data-quality dimensions and thresholds are defined by use case.
- DQ SLOs appear as explicit components of data-product health.
- Service availability and publication reliability are measured separately from data timeliness.
- Discoverability and access lead time are measured.
- Contract stability and breaking changes are tracked.
- Support and incident response expectations are defined.
- Security, privacy and compliance controls are visible.
- Adoption and consumer outcomes are measured.
- Lifecycle controls cover versioning, deprecation and retirement.
- No single DQ percentage is used as the data-product health score.

# Appendix F. Source Notes and Evidence Base

The references below are the evidence base of the original research report. They are grouped by role rather than alphabetically.

## **S01 ISO/IEC 25012:2008 - Data quality model**

<https://www.iso.org/standard/35736.html>

Why it matters: Defines the general structured-data quality model and its characteristics.

## **S02 ISO/IEC 25024:2015 - Measurement of data quality**

<https://www.iso.org/standard/35749.html>

Why it matters: Defines measurement approaches for data-quality characteristics and ties acceptable values to context and user needs.

## **S03 Wang & Strong - Beyond accuracy: What data quality means to data consumers**

[https://web.mit.edu/tdqm/www/tdqmpub/beyondaccuracy\\_files/beyondaccuracy.html](https://web.mit.edu/tdqm/www/tdqmpub/beyondaccuracy_files/beyondaccuracy.html)

Why it matters: Establishes a consumer-oriented view of data quality across intrinsic, contextual, representational and accessibility dimensions.

## **S04 Dehghani - Data Mesh Principles and Logical Architecture**

<https://martinfowler.com/articles/data-mesh-principles.html>

Why it matters: Defines data as a product and describes product composition, ownership, discoverability, understandability and operational qualities.

## **S05 Open Data Product Standard**

<https://github.com/bitol-io/open-data-product-standard/blob/main/docs/README.md>

Why it matters: Shows the broader data-product definition, including management, support, context, teams and authoritative definitions.

## **S06 Open Data Contract Standard**

<https://github.com/bitol-io/open-data-contract-standard>

Why it matters: Separates data quality from schema, support, roles, service levels and infrastructure inside the producer-consumer contract.

## **S07 ISO 19157-1:2023 - Geographic information - Data quality**

<https://www.iso.org/standard/78900.html>

Why it matters: Treats data quality as evidence for conformance to a broader data-product specification.

## **S08 ISO/IEC 25010:2023 - Product quality model**

<https://www.iso.org/standard/78176.html>

Why it matters: Used only as a structural analogy for assessing the whole ICT product, not as a data-product-quality standard.

## **S09 EU GDPR, Regulation (EU) 2016/679**

<https://eur-lex.europa.eu/eli/reg/2016/679/oj/eng>

Why it matters: Separates accuracy obligations from security and accountability obligations.

### **S10 EU AI Act, Regulation (EU) 2024/1689**

<https://eur-lex.europa.eu/eli/reg/2024/1689/oj/eng>

Why it matters: Shows that data quality is one governance concern among broader lifecycle and risk controls for high-risk AI.

### **S11 Basel Committee - BCBS 239**

<https://www.bis.org/publ/bcbs239.htm>

Why it matters: Connects data accuracy, completeness and timeliness with aggregation capability, reporting reliability and operational decision needs.

### **S12 Public Health England - delayed reporting of COVID-19 cases**

<https://www.gov.uk/government/news/phe-statement-on-delayed-reporting-of-covid-19-cases>

Why it matters: Provides a concrete example of DQ effects caused by an end-to-end product delivery failure.

### **S13 USGS - Using the Landsat Level-1 Data Product**

<https://www.usgs.gov/landsat-missions/using-usgs-landsat-level-1-data-product>

Why it matters: Shows a product-oriented pattern where calibrated data are delivered with metadata and documented transformation context.

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## ABOUT THE AUTHOR

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## AI Product Leadership from Strategy to Working Systems

Dr. Jarkko Moilanen works across AI products, data products, open standards and product portfolio management. His work connects business objectives with working AI systems, agentic architectures, machine-readable product definitions and the operating models required to manage them.

He ignited the Open Data Product Specification initiative, which later expanded into today's Linux Foundation Open Data Products standards family. The family now covers individual product definitions, catalogs and portfolios, graph relationships, shared vocabulary and workflow contracts for human, platform and AI-agent use.

This white paper separates external evidence from the author's synthesis. Its central proposition is practical: data quality and data product quality should be separated, connected and managed together.

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